



To: HHS Health Sector AI RFI
From: GeneDx

February 21, 2026

II. Solicitation of Public Comments

GeneDx appreciates the opportunity to provide input on how the Department of Health and Human Services (HHS) can accelerate the effective adoption of artificial intelligence (AI) to propel the American health care system forward. As a leader in delivering improved health outcomes through genomic insights, we are applying AI to accelerate and enhance genetic diagnostics. AI is shaping the future of medical genetics, and we are and we are leveraging our rich data through AI-informed tools to bring answers to more patients, reduce manual workflows, and scale precision medicine for all patients.

AI supports our business in three main ways. First, AI helps unlock GeneDx's full potential to offer more diagnoses to more patients. For example, we enhance the power of our interpretation platform with Multiscore, an advanced AI-powered decision support tool designed to revolutionize genetic analysis by improving diagnostic efficiency, streamlining workflows, and enhancing clinical insights. The tool helps clinical geneticists focus on the most relevant genes during their analysis. Second, AI helps us drive increased speed and accuracy across our operations. Last year, we announced our acquisition of Fabric Genomics, a leader in AI interpretation – combining Fabric Genomics' AI-driven interpretation platform with GeneDx's unparalleled rare disease data set to make patient genome interpretation faster, more scalable, and more impactful. Third, AI enables our entire enterprise to operate more efficiently and effectively. We also leverage Natural Language Processing, Machine Learning, and Large Language Models to drive growth and efficiency wherever possible, including as we transition ourselves to the Lab of the Future.

As GeneDx works to build the Lab of the Future, we are designing a laboratory optimized for continuous improvement—harnessing automation, robotics, and AI-driven algorithms to increase throughput, quality, and consistency. Propelled by one of the world's most comprehensive and clinically curated genomic and phenotypic datasets, and supported by our expert team of geneticists, this model enables a 24/7, highly scalable operation capable of growing from approximately 100,000 patients per year to more than one million, while delivering faster and more accurate answers. This approach allows us to scale efficiently, unlock economies of scale, and help ensure genomic diagnostics are accessible to more patients.

GeneDx applauds HHS for seeking engagement on how AI can improve patient and caregiver experiences and outcomes, reduce provider burden and improve quality of care, and reduce health care costs for consumers and government. We particularly appreciate HHS' focus on future-looking responses, as GeneDx works to build the Lab of the Future.



Our comments below focus on strategies to support the safe and effective clinical adoption of AI. Thank you for soliciting our input.

Specific Questions

In addition to the general requests for information above regarding AI regulation, reimbursement, and research & development, HHS seeks input on the following specific questions:

1. What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

GeneDx appreciates the importance of understanding barriers to private sector innovation in AI-enabled health care. Given the breadth and complexity of this topic, GeneDx would welcome the opportunity to engage further with HHS through future listening sessions, workshops, or other multi-stakeholder forums.

2. What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.

To incentivize the effective use of AI in clinical care, HHS could consider the following:

- Clarifying risk-based pathways that differentiate software as a medical device (SaMD) from non-device clinical support to reduce unnecessary regulatory burdens, while prioritizing safety.
- Developing reimbursement frameworks for clinically validated AI tools to boost innovation.
- Providing regulatory safe harbors and pilot programs for innovative AI integration.
- Endorsing continued delays in scheduled payment reductions to the Clinical Laboratory Fee Schedule under the Protecting Access to Medicare Act to ensure adequate reimbursement rates and maintain patient access to high-quality diagnostic services, particularly for individuals and families affected by suspected rare disease.

3. For non-medical devices, we understand that use of AI in clinical care may raise novel legal and implementation issues that challenge existing governance and accountability structures (e.g., relating to liability, indemnification, privacy, and security). What novel legal and implementation issues exist and what role, if any, should HHS play to help address them?

GeneDx recognizes that liability, indemnification, privacy, and accountability frameworks for AI-enabled tools in clinical care are evolving. We believe further multi-stakeholder engagement is necessary before prescribing specific policy solutions and would welcome continued dialogue with HHS on these issues.



4. **For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (e.g., contracts, grants, cooperative agreements, and/or prize competitions)?**

GeneDx supports continued exploration of AI evaluation methods that emphasize clinical validity, transparency, robustness, and appropriate human oversight. We encourage HHS to continue convening experts to advance best practices in this area.

5. **How can HHS best support private sector activities (e.g., accreditation, certification, industry-driven testing, and credentialing) to promote innovative and effective AI use in clinical care?**

Certification to bolster credibility and grants to fund innovation would help promote innovative and effective AI use in clinical care, as well as a mechanism to share best practices across companies in the health care sector. This should include standards for the diversity of datasets that are leveraged to train AI-informed models; and standards for incorporating evidence generated from AI-informed models into variant classification frameworks.

6. **Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?**

AI tools deployed in genomic testing have exceeded performance and cost expectations. Much of genetic interpretation relies on analyzing large datasets, presenting an opportunity for the use of AI tools, designed to find patterns in the setting of highly complex data, and integrate multiple data types. AI can be leveraged to derive insights from cumulative testing datasets and provide predictions of patients' genetic conditions. For example, GeneDx's Multiscore decision support tool prioritizes genes in exome and genome sequences by ranking them based on alignment with a patient's clinical presentation. Incorporating phenotypic similarities into gene ranking criteria, allows GeneDx to better serve patients whose symptoms may not fit textbook descriptions, and enabling the identification of a definitive or possible molecular diagnosis even in the setting of atypical or expanded clinical presentations. This approach enables faster, more accurate identification of genetic conditions, providing a scalable solution for clinical interpretation embedded directly within GeneDx's platform, and reducing the time required for manual review by analysts, ultimately shortening the overall turnaround time for delivering results.

As the number of patients who could benefit from genomic sequencing continues to grow, AI-enabled tools can help extend the reach of the limited genetics workforce and promote more equitable access to timely, accurate diagnoses.

Although one in ten people live with a rare disease, many go undiagnosed, untreated, and unsupported. On average, finding the correct diagnosis can take five years, more than 16 medical



tests, and \$10,000 in health care costs. By enabling earlier diagnosis and treatment, novel AI tools in genomics offer significant potential to improve health care outcomes, give new insights on quality, and help reduce costs.

7. Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?

Chief Medical Officers and Chief Information Officers have the most influence on the adoption of AI for clinical care. GeneDx also has a Director of AI to accelerate adoption and innovation. Additionally, Dr. Thomas Fuchs, Chief AI Officer at Eli Lilly and Company, serves on our board of directors. Fuchs has extensive experience scaling AI and machine learning infrastructure and algorithms for practical, high-impact applications in diagnostic medicine.

8. Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.

Interoperability would widen market opportunities, fuel research, and accelerate the development of AI for clinical care, as AI is only as powerful as the data it is trained on. Leveraging large, diverse, and unbiased genomic data in a responsible and expert-informed manner would help extend the boundaries of knowledge and find answers for more patients. Population frequency variant data or “synthetic patients,” as leveraged for DNA sequence models, protein-sequence models, and knowledge-based models of disease are also essential.

9. What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?

With the adoption of AI in clinical care, patients and caregivers hope to see shorter diagnosis times and improved access to high-quality, affordable genomic diagnostics and downstream precision care. At the same time, many have concerns about the privacy and confidentiality of health and genomic data, as well as transparency in AI-driven decision-making. For this reason, we prioritize strict data confidentiality and a human-in-the-loop approach – deploying AI in a human-centric way to ensure quality of care for patients. The other concern is the quality and characteristics of the data being leveraged. Attention must be paid to train AI tools with data that are diverse to avoid perpetuating biases in the medical and genomic care spaces.

10. Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?

- a. Are there published findings about the impact of adopted AI tools and their use clinical care?**
- b. How does the literature approach the costs, benefits, and transfers of using AI as part of clinical care?**

HHS should prioritize the potential of AI in genomic sequencing.



An increasing number of clinical guidelines, payor policies, and, more recently, legislation, support GS and ES as the standard of care, indicating that the number of patients who could benefit from this testing will outpace the capacity of the geneticist workforce.

While there are many ways that AI can enhance the business of genetic testing, such as automating billing and for customer support, AI's potential in variant classification and interpretation is particularly compelling. AI uses within medical genetics include automating aspects of the clinical workflow, providing predictions of patients' genetic conditions, and deriving insights from the cumulative testing dataset, including uncovering new gene–disease relationships.

The workflows of a genetics laboratory include analysis and extraction of features such as HPO terms from clinical documents, review of the genetic literature, assignment of American College of Medical Genetics (ACMG) criteria to variants (Richards et al. [2015](#)), and generation of a test report to communicate results to patients and clinicians. In all these use cases, AI can provide solutions to perform tasks more reproducibly, more thoroughly, and faster. For example, conventional or generative AI can be used to extract structured clinical information from clinical documents (Fu et al. [2024](#); Srivastava et al. [2024](#)), mine the genetics literature for evidence attributable to the variants (Lyu et al. [2024](#); Wei et al. [2018](#)), and pull together relevant test information and information from the literature to draft a genetic test report. AI has been used for similar tasks within other knowledge-intensive fields such as legal (e.g., <https://www.clio.com>) and software (e.g., <https://www.cursor.com>) realms. The summarization and synthesis of genetic findings from many publications and internal and external cases can be enhanced by agentic workflows that generate knowledge-grounded first drafts of passages to consider for inclusion in the final patient report.

Identifying relevant variants and determining their pathogenicity is another extremely important part of the workflow that can be powered by AI (Aradhya S. et al. 2023). This is a process of whittling thousands of potential variants down to one or several that are responsible for the patient's clinical presentation. There has been great advancement in DNA sequence models like ESM-var (Meier et al. [2021](#)) and protein-sequence models such as AlphaMissense (Cheng et al. [2023](#)) that can predict pathogenicity of genetic variants. Additionally, knowledge-based models of disease such as SHEPHERD (Alsentzer et al. [2024](#)) provide powerful tools for recommending the causative variant or gene in a particular patient's case. Many of these methods rely on clever strategies to find data upon which to train, for example, population frequency variant data in the case of AlphaMissense or “synthetic patients” for SHEPHERD. Large commercial laboratories have robust patient datasets of ES and GS data paired with detailed clinical information for patients suspected of rare disease, which can also be used to train highly effective models of variant pathogenicity.

In addition to helping to solve a single genetic case, AI can leverage large public and commercial genetic datasets that couple genotype and phenotype information to extend the boundaries of knowledge and find answers for more patients. AI models can integrate the mountains of



genomic data to find new gene-disease candidates, help understand pathogenic non-coding variants, and generate models of polygenic risk (Greene et al. [2017](#); Wells et al. [2019](#)).

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